

High-order local non-reflecting boundary conditions: a review[☆]

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Received 1 September 2003; received in revised form 5 October 2003; accepted 9 October 2003

Abstract

A common method for numerically solving wave problems in unbounded domains is based on truncating the infinite domain via an artificial boundary B , thus defining a finite computational domain, and using a special non-reflecting boundary condition (NRBC) on B . Low-order local NRBCs have been constructed and practiced since the 1970s. Exact non-local NRBCs were introduced in the 1980s. Only recently *high-order local* NRBCs have been devised. These NRBCs, despite being of an arbitrarily high-order, do not involve high derivatives owing to the use of specially defined auxiliary variables. This paper reviews the latter approach, explains its advantages compared to previous approaches, and discusses the different schemes which have been proposed in this context.

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Keywords: Non-reflecting boundary conditions; Absorbing boundary conditions; Finite elements; Finite differences; Auxiliary variables; High-order

1. Introduction

Waves in unbounded media occur in many fields of application, such as acoustics, aerodynamics, solid geophysics, oceanography, meteorology and electromagnetics. Numerical methods for such problems have been developed since the 1970s [1]. The main four types of methods that have emerged are: boundary integral methods, infinite element methods, absorbing layer methods and non-reflecting boundary condition (NRBC) methods.

The latter family of methods can be described as follows. The infinite domain is truncated via an artificial boundary B , thus dividing the original domain into a finite computational domain Ω and a residual infinite domain D . A special boundary condition is imposed on B , in order to complete the statement of the problem in Ω (i.e., make the solution in Ω unique) and, most importantly, to ensure that no (or little) spurious wave reflection occurs from B . This boundary condition is called a NRBC, although a few other names, such as absorbing boundary condition, are often used too [2]. The problem is then solved numerically in Ω , say by the finite element (FE) or finite difference (FD) method.

[☆] Partially supported by the Technion's R&M Rosenthal Aerospace Engineering Research Fund and by the US National Research Council (NRC).

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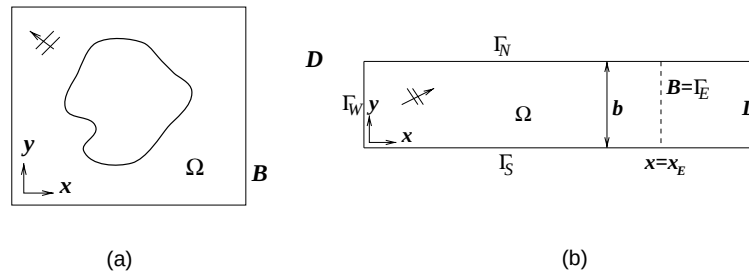


Fig. 1. Setup for the NRBC method: (a) an exterior scattering problem; (b) a semi-infinite wave-guide problem.

The setup is illustrated in Fig. 1. Fig. 1(a) pertains to an exterior problem outside of a scatterer or an obstacle. The artificial boundary B has a rectangular shape in the figure, although sometimes a smooth shape (like a circle in two dimensions or a sphere in three dimensions) is preferred. Fig. 1(b) describes a semi-infinite wave-guide problem. In the example shown, $B = \Gamma_E$ is a cross-section of the wave-guide which constitutes the east side of Ω .

The problem of constructing a “good” NRBC may seem very simple at first sight. After all, all that is needed is a boundary condition that lets all the waves that impinge on the boundary go out freely without giving rise to spurious reflection. However, this apparent simplicity is deceiving. It turns out that it is very difficult to find a single NRBC that is sufficiently general and which after discretization leads to a scheme which is stable, accurate, efficient and easy to implement. This explains the fact that in the last 25 years or so, much research has been devoted to developing improved NRBCs. This problem has been the subject of several conference sessions and special journal issues [3–5]. There is still no consensus on the optimal NRBC. See the review papers [6–9] for more details.

The main milestones in the history of NRBCs are as follows:

- (1) Till the late 1970s the use of the Sommerfeld-like NRBC has dominated the field [1,2]. However, soon it became clear that this NRBC provides a very crude approximation. In fact, today it is thought of as a “zero-order” boundary condition.
- (2) From the late 1970s to the mid-1980s, a few improved low-order NRBCs have been proposed. Some of them became well-known, e.g., the Engquist–Majda NRBCs [10] and the Bayliss–Turkel NRBCs [11]. The second-order NRBCs in these two sequences have especially become popular, and are still commonly used today.
- (3) The late 1980s and early and mid-1990s have been characterized by the emerging of exact non-local NRBCs like those based on the Dirichlet-to-Neumann (DtN) map [12,13] and on the Difference Potential Method (DPM) [14,15].
- (4) In the mid-1990s the perfectly matched layer (PML) [16] was invented.
- (5) Since the mid-1990s *high-order local* NRBCs have been developed. These are the subject of the present paper.

It should be mentioned that some other methods for wave problems in unbounded domains have been developed in parallel to these NRBC methods. Particularly important are boundary integral methods (see, e.g., [17]) and infinite element methods (see, e.g., [18,19]).

This paper is a review of work that has been done on *high-order local* NRBCs since the mid-1990s. Sequences of increasing-order NRBCs have been available before (e.g., the Bayliss–Turkel conditions [11] constitute such a sequence), but they had been regarded as impractical beyond second- or third-order from the implementation point of view. Only since the mid-1990s, *practical* high-order NRBCs have been devised. We explain the advantages of this approach compared to previous approaches, with the exception of the PML method which is outside the scope of this paper. Then we discuss seven different high-order NRBC schemes that have been proposed.

2. High-order local NRBCs: the general approach

First, let us define what we mean by “high-order NRBCs”. By this term we mean a *sequence* of NRBCs with the following two properties:

- The NRBCs in this sequence are of *increasing accuracy*, so that their “order” is associated with their level of accuracy. Different measures of accuracy are possible.
- The NRBCs must be *practically* implementable for an arbitrarily high-order.

The second property is crucial. For a NRBC to be regarded as high-order, we should be able to implement it by an actual computer code such that the order is simply an input parameter. This property is not possessed by the classical sequences of NRBCs. For example, consider the Bayliss–Turkel sequence of NRBCs [11] for the scalar wave equation in three dimensions with a spherical artificial boundary B of radius R :

$$\left(\prod_{j=1}^J \left(\frac{1}{C} \partial_t + \partial_r + \frac{2j-1}{R} \right) \right) u = 0 \quad \text{on } B. \quad (1)$$

Here C is the wave speed. Although J in (1) may be theoretically large, this NRBC, at least in its original form (1), cannot be implemented up to an arbitrarily high-order due to the *high derivatives* appearing in it. As (1) shows, the J th-order condition involves a product of J first-order normal and time derivatives and thus leads to J th-order differential operators whose discrete representation for an arbitrarily high J is not practical.

A high-order NRBC is not necessarily *exact*. We define an exact NRBC as a high-order NRBC whose associated error approaches zero as the order goes to infinity. In other words, an exact NRBC is one that is converging. The way to define such convergence is the following. Consider the exact solution of two problems: the first is the original problem in the infinite domain, and the second is the problem in the truncated domain Ω , with the NRBC applied on B . Then the distance δ , in some reasonable norm, between the two solutions in Ω may serve as an error measure. Now, if a high-order NRBC satisfies the condition that for *any* given problem of the class under consideration the error measure δ of the NRBC approaches zero as the order J goes to infinity while B is held fixed then the NRBC is called exact or converging. High-order NRBCs which are not exact are typically asymptotic in nature.

High-order NRBCs may be divided into two categories: non-local and local ones. This division is related to the price one has to pay to construct a sequence of high-order conditions. Fig. 2 illustrates the tradeoff. With a *non-local* condition all the degrees of freedom on the boundary are coupled. In FE applications this means that the usual FE architecture of doing all the computations on the element level cannot be carried out as far as the NRBC calculations are concerned; these calculations must usually be done on the global level. In addition, on the algebraic level the coupling of the boundary nodes manifests itself in that the coefficient matrix (the stiffness matrix in FE formulations) includes a full block. These attributes have been regarded as disadvantages of exact non-local NRBCs

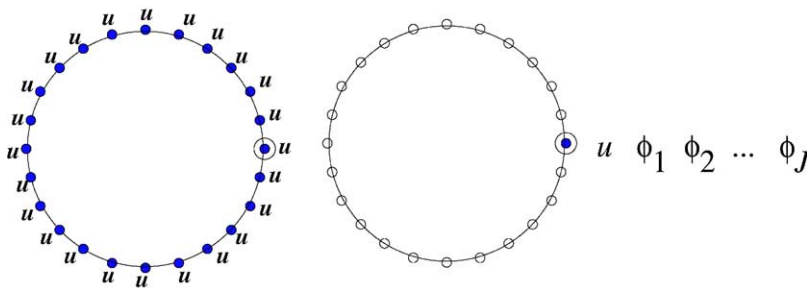


Fig. 2. Non-local vs. local high-order NRBCs: with a non-local condition (left) all the degrees of freedom on the boundary are coupled; with a local condition (right) each node on the boundary is associated with a number of degrees of freedom (auxiliary variables) in addition to the physical ones.

such as the DtN condition, although various arguments may be given to counter this claim; see, e.g., the discussion in [7].

On the other hand, with a *local* high-order NRBC each node on the boundary interacts only with its immediate neighbors, as usual in FE and FD calculations. All the FE calculations may be carried out on the element level. However, each node on the boundary is associated with a number of degrees of freedom in addition to the physical ones. These are auxiliary variables which essentially enable us to eliminate the high derivatives from the NRBC formulation. It is felt that the price entailed in the use of auxiliary variables is more convenient compared to the coupling that comes with the non-locality, at least in the FE context. After all, the notion of adding more degrees of freedom for better accuracy is standard and appears in various forms in FE methodology. It seems unreasonable to expect that a high-order NRBC can be constructed such that it is compatible with FE architecture yet does not involve additional degrees of freedom. We also remark that in most high-order local NRBC schemes the computational effort increases only *linearly* with the order of the NRBC, which seems like a very moderate price to pay for the additional accuracy obtained.

3. Review of high-order local NRBCs

Seven different high-order local NRBCs have been proposed in the last 10 years. They are summarized in Table 1. We now briefly comment on each of them.

Collino [20] can be regarded as the pioneer of high-order local NRBCs, since (to the best of the author's knowledge) he devised the first such NRBC in 1993. We shall summarize his approach here, with a slight change in the notation.

Consider the two-dimensional wave equation

$$\partial_t^2 u - C^2 \nabla^2 u = 0. \quad (2)$$

We assume that the domain includes a straight artificial boundary B (one side of a wave-guide or of a rectangular computational “box”), and we let x be the direction normal to B and y the direction tangent to it. If we Fourier transform (2) in x , y and t (or formally replace ∂_x , ∂_y and ∂_t by $-ik_x$, $-ik_y$ and $-i\omega$, respectively) we get the standard dispersion relation

$$-\frac{\omega^2}{C^2} + k_x^2 + k_y^2 = 0, \quad (3)$$

Table 1
High-order local NRBCs

Authors	Refs.	PDE	Configuration	Discr.	Principle
Collino	[20]	Scalar wave equation	2D Cartesian	FD	Rational approximation
Grote and Keller ^{a,–c}	[21–26]	Scalar wave equation; elasticity; Maxwell	3D spherical	FD, FE	Spherical harmonics
Hagstrom and Hariharan ^d	[27,28]	Scalar wave equation; Maxwell equations	2D cylindrical and 3D spherical	FD	B-T operators
Guddati and Tassoulas	[29]	Scalar wave equation	2D Cartesian	FE	Continued fraction
Rowley and Colonius	[30]	Linear hyperbolic systems ^e	2D Cartesian	FD	Discrete dispersion
Givoli and Patlashenko	[31,32]	Helmholtz equation	2D cylindrical and 2D wave-guide	FE	Localized DtN
Givoli et al.	[33–36]	Dispersive wave equation; LSWE ^f	2D Cartesian	FD, FE	Higdon operators

^a Sofronov [37] independently published a similar scheme in the Russian literature.

^b This NRBC is not purely local: it is local in time but non-local in space on B .

^c See a later derivation and FE implementation by Thompson and Huan [38,39].

^d See a later derivation and FE implementation by Huan and Thompson [40,41].

^e Test case: linearized Euler equations.

^f LSWE: linearized shallow water equations.

where ω is the time-frequency and k_x and k_y are the x and y wave numbers. From (3) we have

$$k_x \bar{u} = \left(\frac{\omega}{C}\right) \sqrt{1 - \frac{k_y^2 C^2}{\omega^2}} \bar{u}, \quad (4)$$

where $\bar{u}$ is the Fourier transform of u in x , y and t . In (4) we have taken the positive root which corresponds to outgoing waves. Now we take the inverse Fourier transform of (4) in the x direction only. This yields

$$\partial_x \bar{u} + i \left(\frac{\omega}{C}\right) \sqrt{1 - \frac{k_y^2 C^2}{\omega^2}} \bar{u} = 0 \quad \text{on } B. \quad (5)$$

Here $\bar{u}$ is the inverse Fourier transform of $\bar{u}$ in x . We denote $z = k_y^2 C^2 / \omega^2$. Then we approximate the irrational function $\sqrt{1 - z}$ appearing in (5) by a rational function, using the Padé approximation:

$$\sqrt{1 - z} \simeq 1 - \sum_{j=1}^J \frac{\beta_j z}{1 - \alpha_j z} \quad \text{for } z \in [0, 1], \quad (6)$$

$$\alpha_j = \cos^2 \left(\frac{j\pi}{2J+1} \right), \quad \beta_j = \frac{2 \sin^2(j\pi/(2J+1))}{2J+1}. \quad (7)$$

We substitute this approximation in (5) to obtain

$$\partial_x \bar{u} + i \left(\frac{\omega}{C}\right) \bar{u} - i \left(\frac{\omega}{C}\right) \sum_{j=1}^J \frac{\beta_j k_y^2}{\omega^2 / C^2 - \alpha_j k_y^2} \bar{u} = 0 \quad \text{on } B. \quad (8)$$

Up to this point the development is similar to that used by Engquist and Majda [10]. However, at this point Collino introduced the *auxiliary variables* (in the transformed plane) $\bar{\phi}_j$ defined by

$$\bar{\phi}_j = \frac{k_y^2}{\omega^2 / C^2 - \alpha_j k_y^2} \bar{u}. \quad (9)$$

With this definition (8) gives

$$\partial_x \bar{u} + i \left(\frac{\omega}{C}\right) \bar{u} - i \left(\frac{\omega}{C}\right) \sum_{j=1}^J \beta_j \bar{\phi}_j = 0 \quad \text{on } B. \quad (10)$$

From (9) we also have

$$\left(\frac{\omega^2}{C^2}\right) \bar{\phi}_j - \alpha_j k_y^2 \bar{\phi}_j = k_y^2 \bar{u} \quad \text{on } B. \quad (11)$$

Finally we take the inverse Fourier transform of (10) and (11) in y and t and obtain

$$\partial_x u + \frac{1}{C} \partial_t u - \frac{1}{C} \sum_{j=1}^J \beta_j \partial_t \phi_j = 0 \quad \text{on } B, \quad (12)$$

$$\frac{1}{C^2} \partial_t^2 \phi_j - \alpha_j \partial_y^2 \phi_j - \partial_y^2 u = 0, \quad j = 1, \dots, J \quad \text{on } B. \quad (13)$$

Eqs. (12) and (13) constitute the Collino NRBC of order J . In total these are $J + 1$ coupled equations on B for the $J + 1$ functions $u, \phi_1, \dots, \phi_J$.

The Collino NRBC has the general structure shared by all the high-order local NRBCs. It involves no high derivatives beyond second-order, and it involves no normal (x –) derivatives of any auxiliary variable ϕ_j . The latter fact is essential since the auxiliary variables “live” only along the boundary B .

We note that (12) looks like a modified Sommerfeld-like condition, whereas (13) looks like a one-dimensional wave equation for ϕ_j on B . The latter fact implies, among other things, that when this NRBC is imposed along a rectangular artificial boundary B which has *corners*, special corner conditions must be used which serve as “boundary conditions” for the wave equation (13). Without such corner conditions the problem would become ill-posed, and this would manifest itself in the loss of stability on the discrete level. Collino [20] proposed such corner conditions, and demonstrated the performance of the whole FD scheme via a numerical example.

It should be remarked that although Collino’s formulation is given in two dimensions (2D), the ideas carry over immediately to three dimensions (3D) as well. The same applies in principle to all the NRBCs whose configuration is indicated as “2D Cartesian” in Table 1.

The Collino NRBC is certainly a high-order condition by the definition given previously. However, it is not always an exact condition; whether or not it is exact depends on the convergence of the Padé series.

The Grote–Keller NRBC [21–26] is also a pioneering contribution in that it was the first exact NRBC which is local in time on a spherical boundary. The existence of such a NRBC was an open question before the first Grote–Keller paper. In fact, an exact time-dependent NRBC that was available before that time is based on the Kirchhoff integral [42,43] and is non-local in time (although the non-locality is “fixed” and does not grow in time). We note that the Grote–Keller condition is inherently 3D, since it is based on the special properties of the spherical harmonics. Moreover, it is believed that no such exact NRBC can be constructed in two dimensions. This is related to the fact that the Green’s function in 2D associated with the wave operator has an infinitely long tail.

The Hagstrom–Hariharan NRBC [27,28] is a reformulation of the Bayliss–Turkel conditions which are given in the 3D case by (1). However, whereas the Bayliss–Turkel NRBC in its original form involves high derivatives which makes it impractical beyond second-order, the Hagstrom–Hariharan NRBC involves no high derivatives owing to the use of special auxiliary variables. The 3D Hagstrom–Hariharan NRBC is exact, just like the Grote–Keller condition, whereas the 2D Hagstrom–Hariharan NRBC is asymptotic.

Thompson and Huan [38–41] investigated and implemented (using FEs) both the Grote–Keller and the Hagstrom–Hariharan NRBCs and discussed the relations between them.

The Guddati–Tassoulas NRBC [29] is based on rational approximation of the dispersion relation, similarly to the Collino NRBC, but the formulation is different and relies on continued fractions. After FE discretization the global system of equations has the form

$$M\ddot{U} + C\dot{U} + KU + R \int_0^t U \, dt = F. \quad (14)$$

Here U is the unknown vector whose entries are the nodal values of u and of all the auxiliary variables ϕ_j (on B). The “damping matrix” C is contributed solely by the NRBC on B since the original problem is undamped. The time-integral term in (14) is non-standard, but turns out not to be at all problematic.

The Rowley–Colonius NRBC [30] is different from all the other NRBCs in that it is designed on the discrete level rather than on the continuous level. This enables the authors to control the spurious reflections resulting from the numerical dispersion error.

The Givoli–Patlashenko NRBC [31,32] is an exact NRBC for the 2D Helmholtz equation. This approach cannot be generalized to the time-dependent case, since it is based on the localization of the non-local DtN map given in the time-harmonic case. It should be remarked that all the other NRBCs, which have been designed for the time-dependent case, can easily be adapted to the time-harmonic case by simply replacing every occurrence of the operator $(1/C)\partial_t$ by $-ik$ (where k is the wave number).

The NRBC proposed by Givoli et al. [33–36] is, as far as the author is aware, the only NRBC that takes physical (as opposed to numerical) wave dispersion into account. Wave dispersion is an important phenomenon occurring in various fields of application like meteorology and oceanography. A prototype equation in this case is the dispersive

wave equation (or Klein–Gordon equation)

$$\partial_t^2 u = C_0^2 \nabla^2 u - f^2 u, \quad (15)$$

where C_0 is the reference wave speed and f is the dispersion parameter. The NRBC devised for this equation in 2D on a straight artificial boundary B normal to x consists of the following:

$$\left(\partial_x + \frac{1}{C_1} \partial_t \right) u = \phi_1, \quad (16)$$

$$\left(\frac{1}{C_j} + \frac{1}{C_{j+1}} \right) \partial_t \phi_j = \phi_{j+1} + \left(\frac{1}{C_j^2} - \frac{1}{C_0^2} \right) (\partial_t^2 + \partial_y^2) \phi_{j-1} - \frac{f^2}{C_0^2} \phi_{j-1}, \quad (17)$$

$$j = 1, \dots, J-1, \quad \phi_0 \equiv u, \quad \phi_J \equiv 0. \quad (18)$$

The ϕ_j are the auxiliary variables defined on B . The parameters C_j are positive constants that may be chosen either by the user or automatically as discussed in [44]. In [33] this NRBC is shown to be exact in some weak sense. Of course, the same NRBC applies to the standard wave equation (2) by simply setting $f = 0$.

The references indicated in Table 1 include numerical examples which demonstrate the performance of the different NRBCs. They also include detailed discussion on issues like well-posedness, numerical stability, computational effort, implementation, etc. The interested reader should consult in these references. All of these NRBC schemes are powerful and extremely useful. The author is confident that high-order local NRBCs are going to dominate the field of artificial boundary treatment for wave problems in the near future.

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